

TABLE 3
FEATURES OF AEM20940 AND SPV1050

Model	Manufacturer	Storage types it can charge	Energy it can harness	Output voltage range	Output current	MPPT	Cold start
AEM20940	e-peas semiconductor	Rechargeable batteries, Capacitors	Thermal energy	2.2-4.5	80mA (max)	Yes	380mV
SPV1050	ST Microelectronics	Rechargeable batteries, Capacitors	Thermal and solar	2.2-5.3V	30-70mA	Yes	0.5V (Boost conversion)

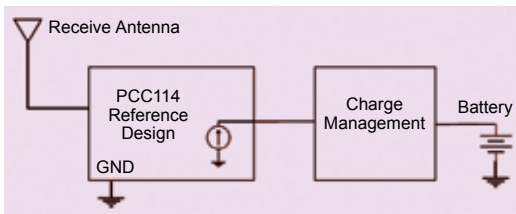


Fig. 4: Reference design of a RF energy harvester

Thermal energy harvesting

Thermal energy harvesting is the process of harnessing heat dissipated by machines or electronic components. Energy harvester

ICs, such as e-Peas AEM20940 or STM SPV1050, are capable of thermal energy harvesting. These devices use thermos-electric generators (TEGs) as the transducers to convert ambient heat energy into electrical energy. Thermal energy harvesting finds usage in applications with a high heat loss, such as in power electronics circuits or electro-mechanical systems. A typical application circuit of e-Peas AEM20940 is shown in Fig. 3. Table 3 lists out the features of energy harvesters from STM and e-peas.

Radio frequency energy harvesters

Radio frequency energy harvesting (RFEH) is a technique of harnessing energy from the electromagnetic (EM) waves produced by sources such as phone stations, TV and radio broadcasting stations, Wi-Fi routers, Bluetooth Low Energy (BLE), etc. As shown in Fig. 4, RF harvesting devices have antennas that receive the EM waves and convert them to DC energy, and charge management steps up the DC voltage to be used for power or charging a battery. Fig. 4 shows an RF energy harvester taken from the PCC114 datasheet.

Radio frequency energy harvester (RFEH) devices like PCC114 from Powercast, AEM40940 from e-peas, and RFD102A from RF Diagnostics capture the electromagnetic vibrations of specific frequencies that are available around us to produce electrical energy. The flow diagram shown in Fig. 5 gives an overview of different modes of AEM40940.

RF power has lower energy density than most other sources, but this method is extremely useful for sensor nodes that are present in isolated areas with availability of no other ambient energies like solar, wind, or thermal. RFEH is commonly used for powering smart wearables, IoT devices, RFID tags, etc. Table 4 compares some com-

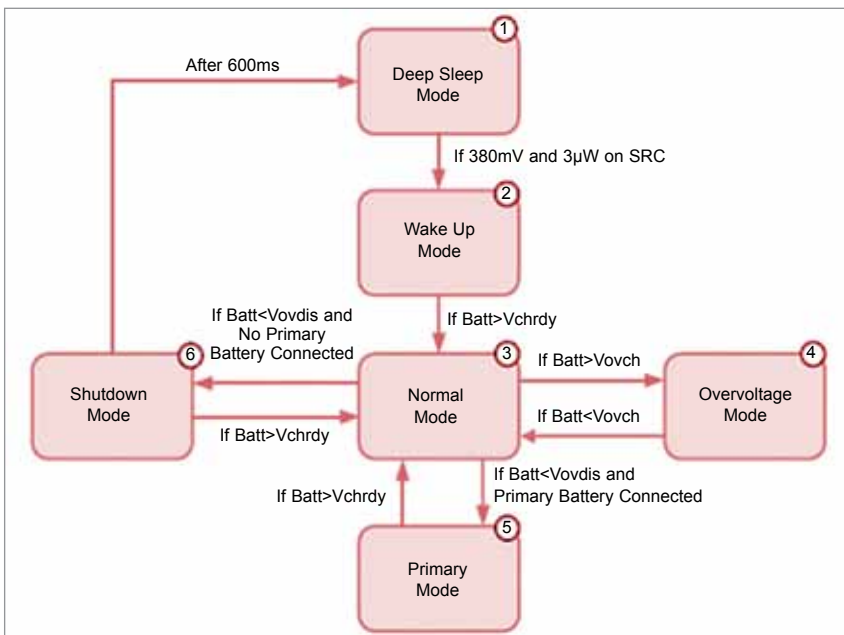


Fig. 5: Modes of AEM40940 from e-peas

TABLE 4
COMPARISON OF e-peas ENERGY HARVESTING ICs

Model	Manufacturer	Storage types	Energy it can harness	Harvesting range	Output voltage range	MPPT
AEM40940	e-peas semiconductor	Li-ion, Capacitor	RF energy harvesting	868-915MHz	50mV-5V	Yes
PCC114	Powercast	Li-ion, Capacitor	RF energy harvesting	10MHz-6GHz	4.3V	Yes
RFD102A	RF Diagnostics	Li-ion, Capacitor	RF energy harvesting	60Hz to 6GHz	4.2V	Yes